



European Energy Analytics

Electricity Generation, Transition and Carbon Emissions

About this report: This report explores electricity generation, demand, energy transition, carbon emissions and installed renewable capacity across selected European countries using monthly energy data.

Countries covered: Austria, Belgium, Czechia, Denmark, Finland, France, Germany, Greece, Hungary, Ireland, Italy, Netherlands, Norway, Poland, Portugal, Romania, Spain, Sweden, Switzerland and the United Kingdom.

Key Analytical Questions

- 01.** How have electricity generation and demand evolved across European countries?
- 02.** How has Europe's electricity generation mix changed over time?
- 03.** How have CO₂ emissions and carbon intensity developed over time?
- 04.** How has installed solar and wind capacity evolved across European countries?

2017



France



Power Generation (TWh)

478,06

Electricity Demand (TWh)

478,03

CO₂ Emissions (MtCO₂)

36,88

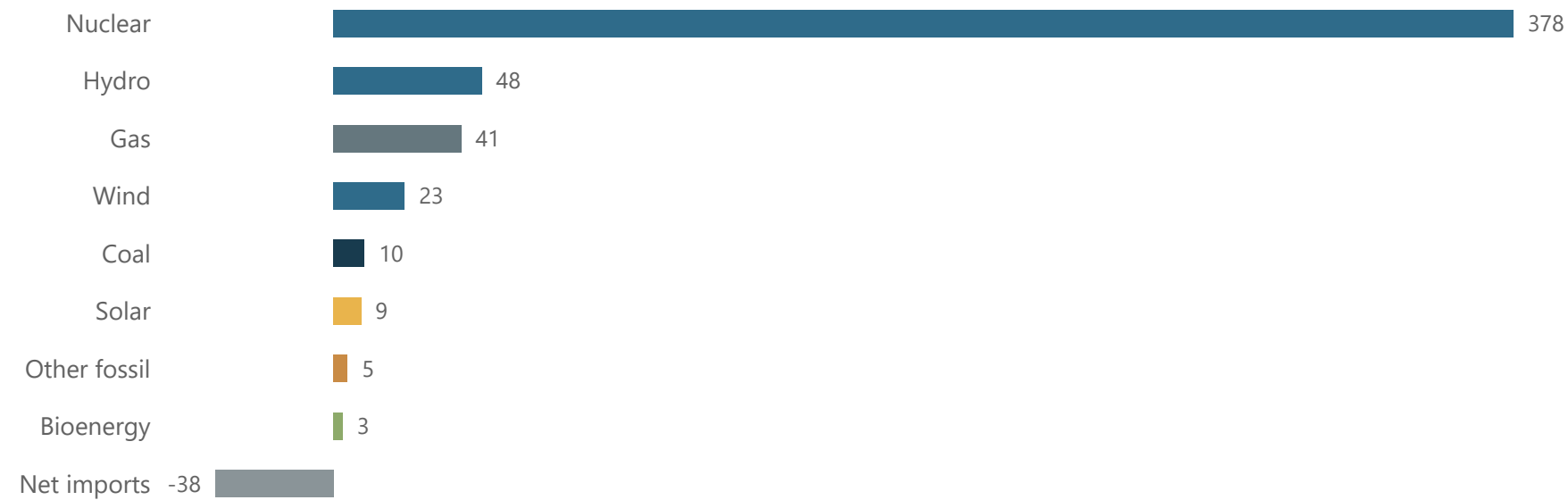
Carbon Intensity (gCO₂/kWh)

69,95

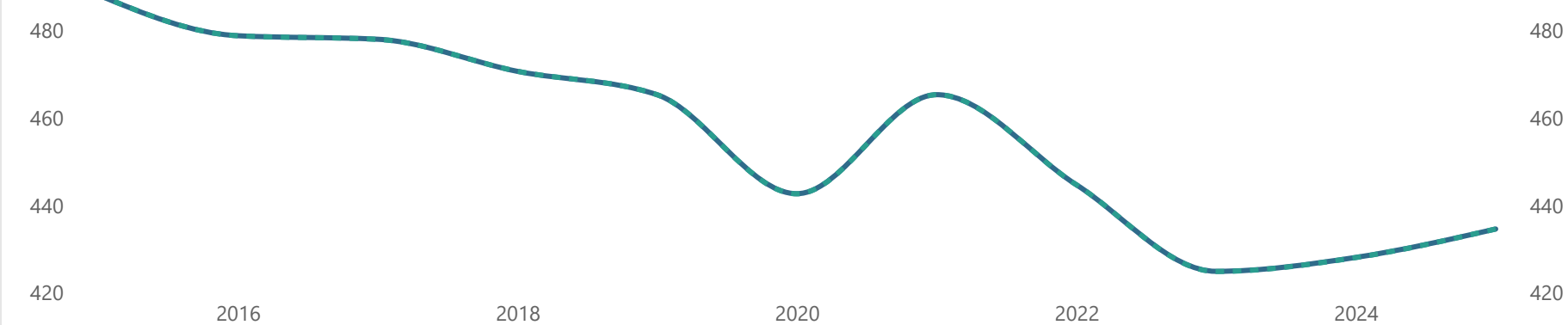
Installed Capacity (GW)

27,81

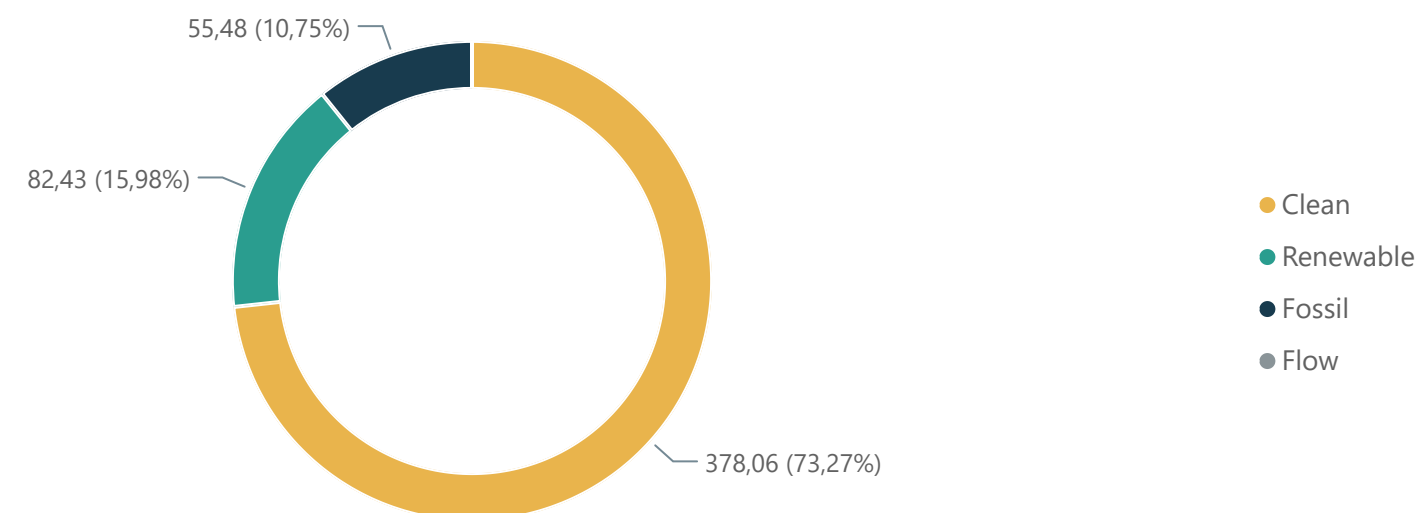
Electricity Generation Mix (TWh)



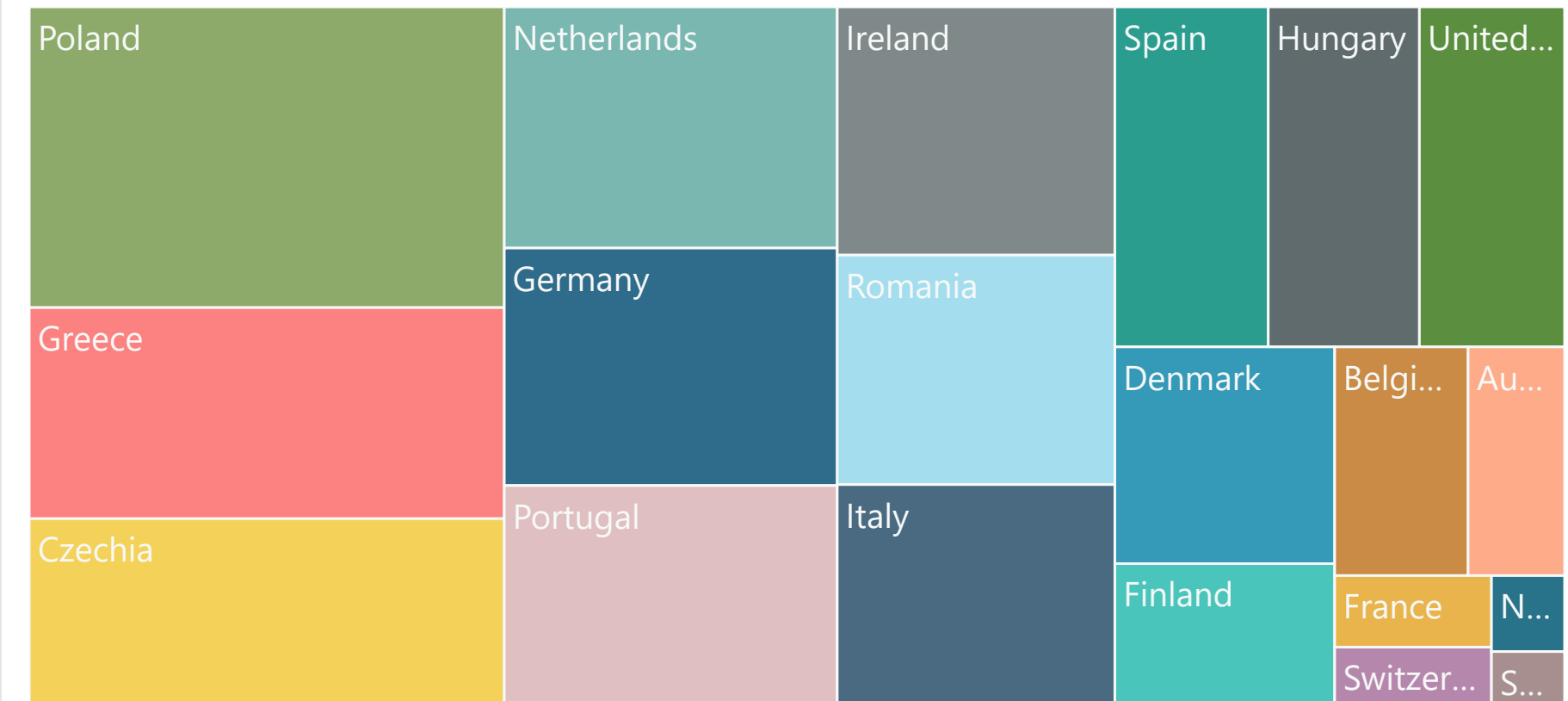
Generation and Total Demand (TWh)



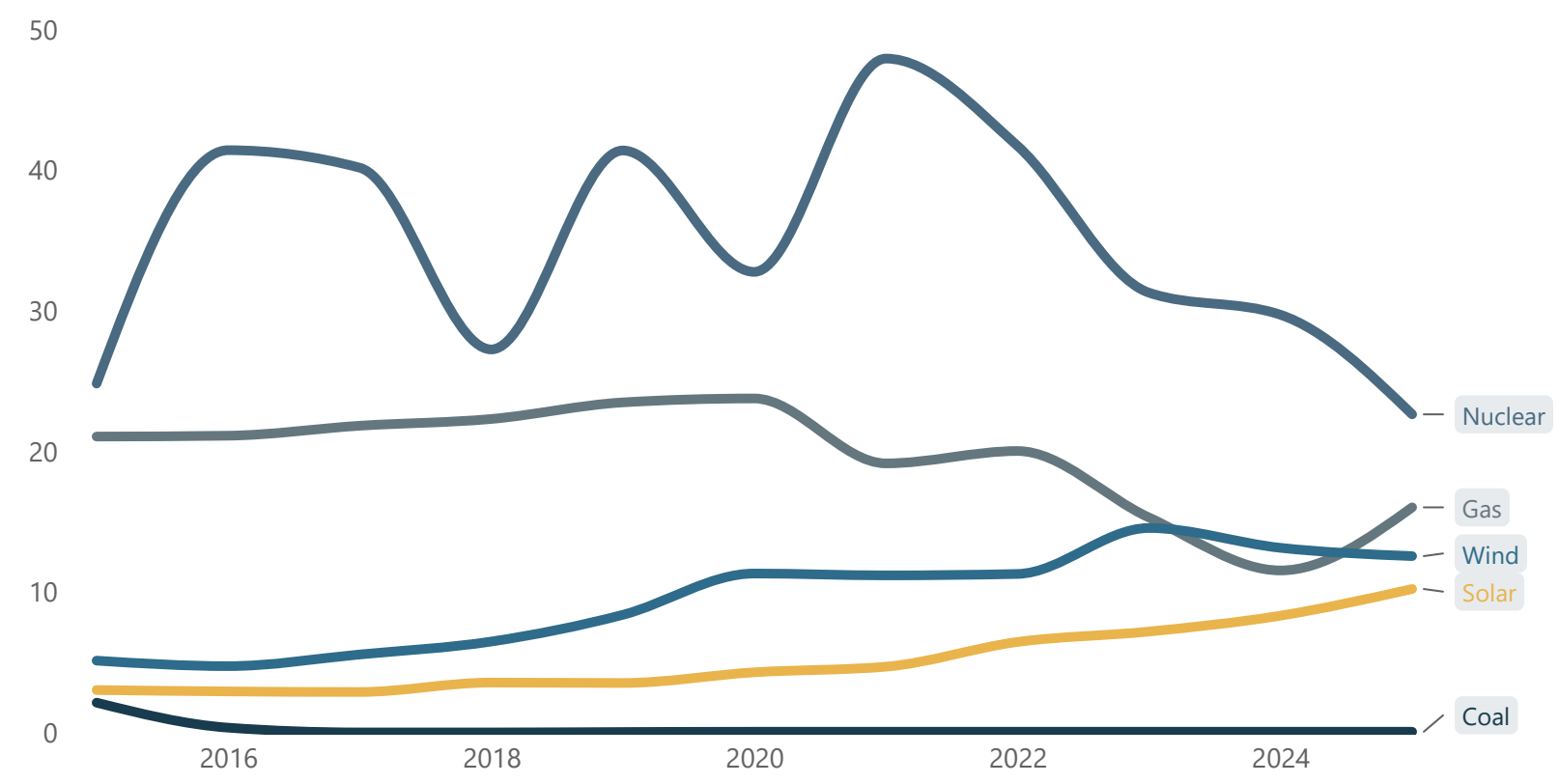
Electricity Generation Mix by Category (TWh)



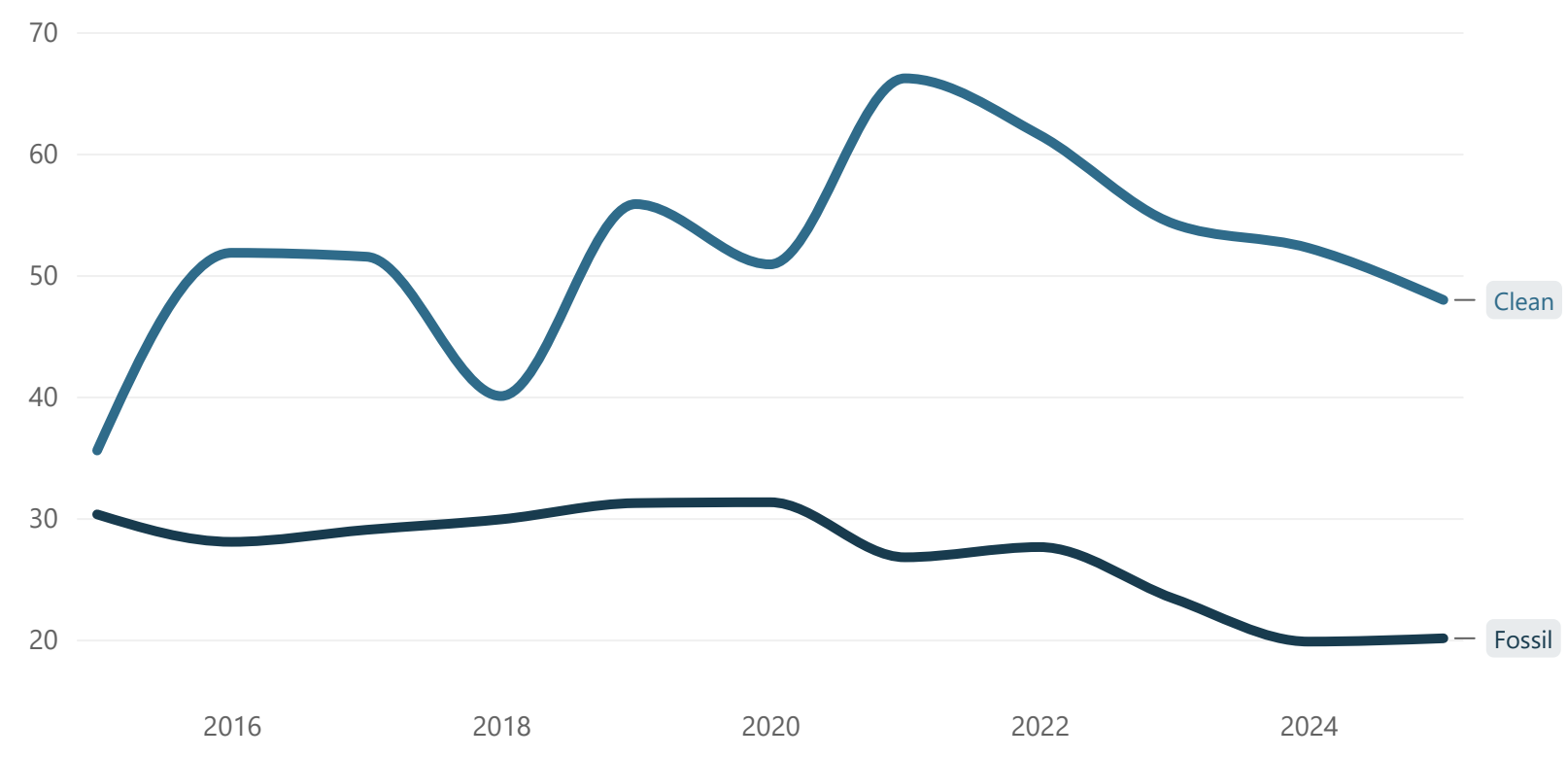
Carbon Intensity by Country (gCO₂/kWh)



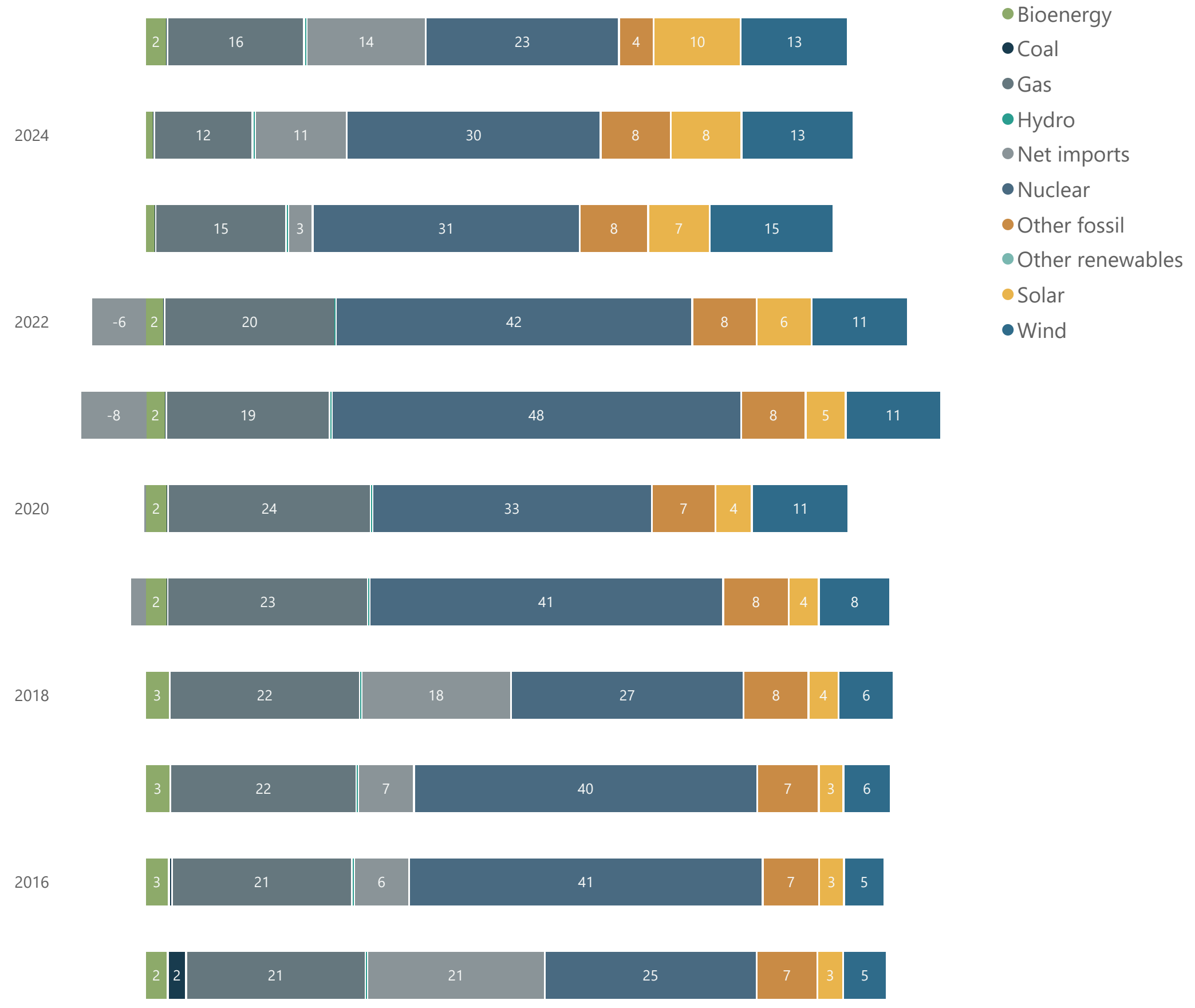
Generation Trends by Energy Source (TWh)



Clean vs Fossil Generation (TWh)



Electricity Generation Mix (TWh)



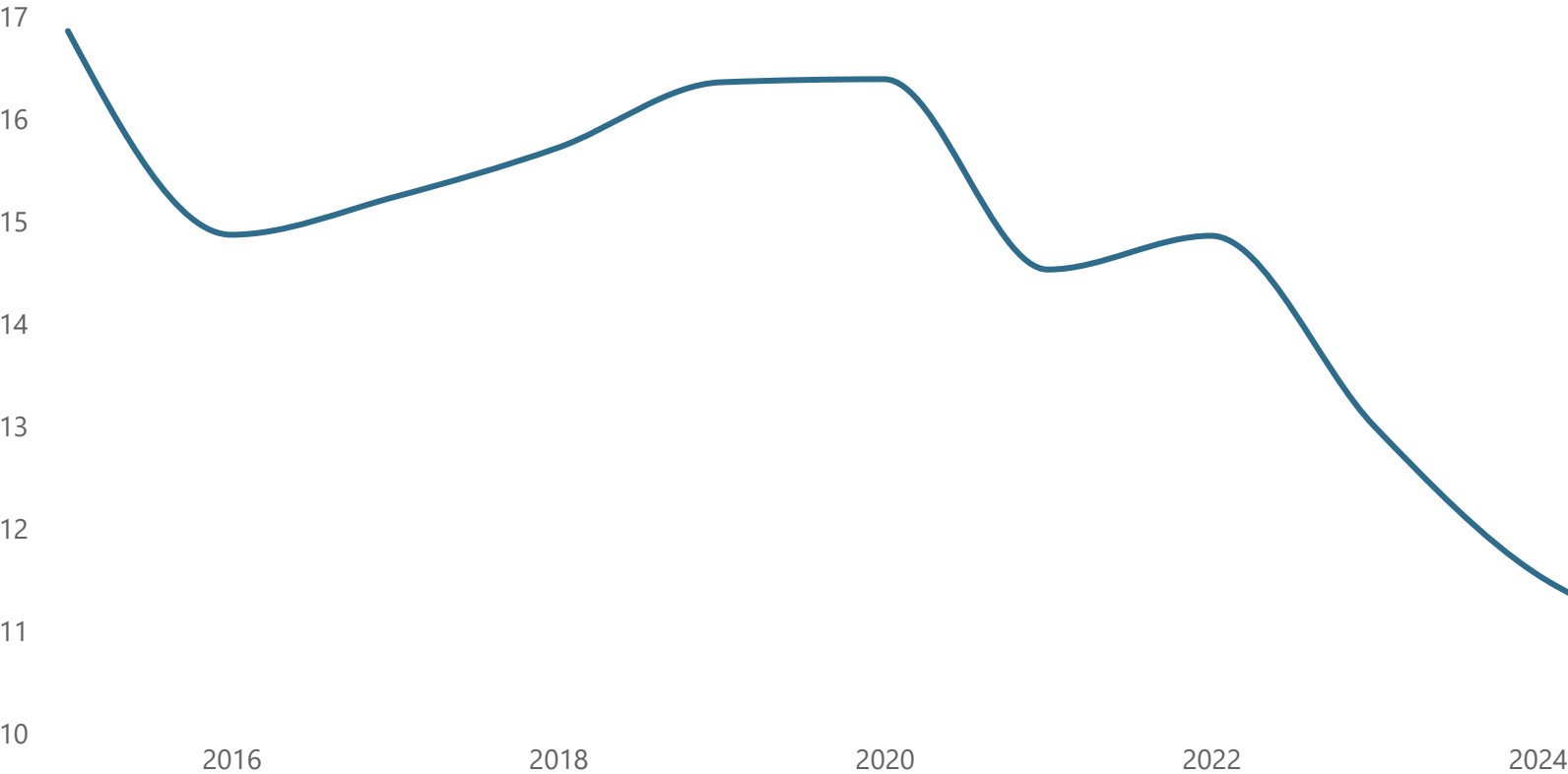
2023



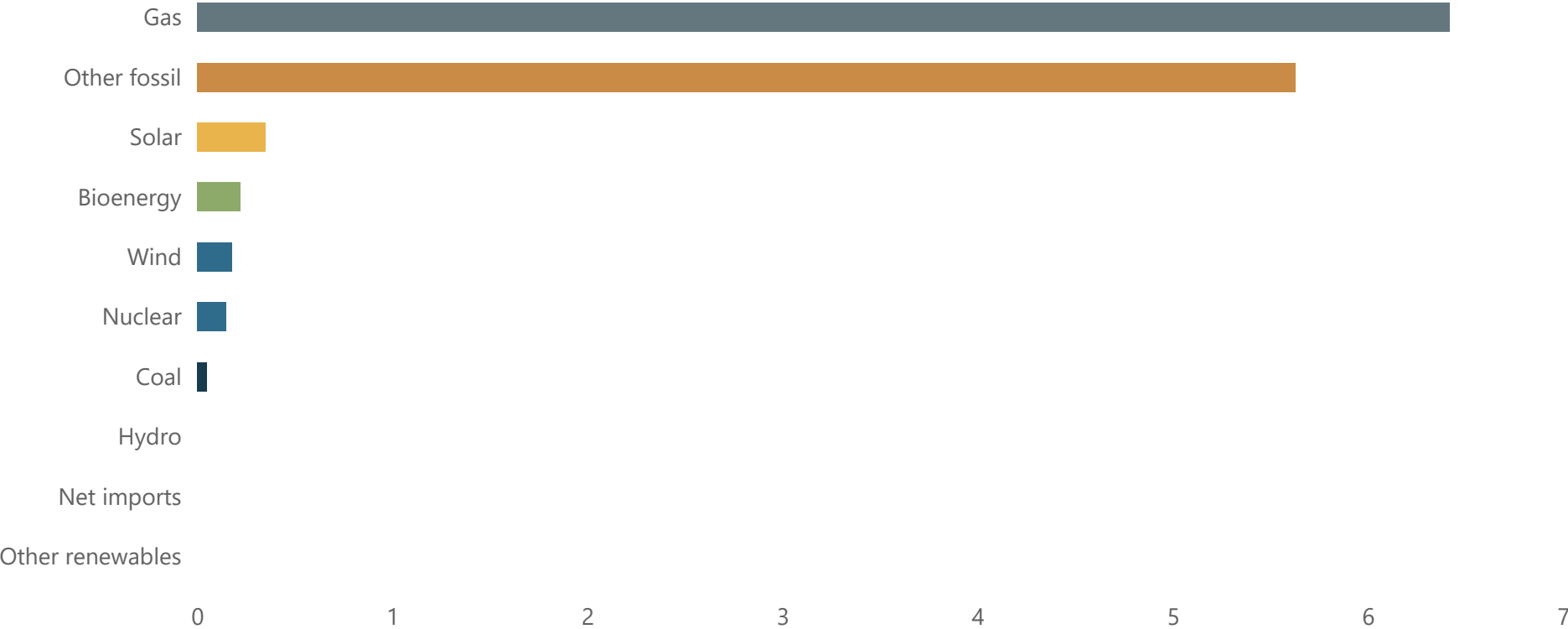
Belgium



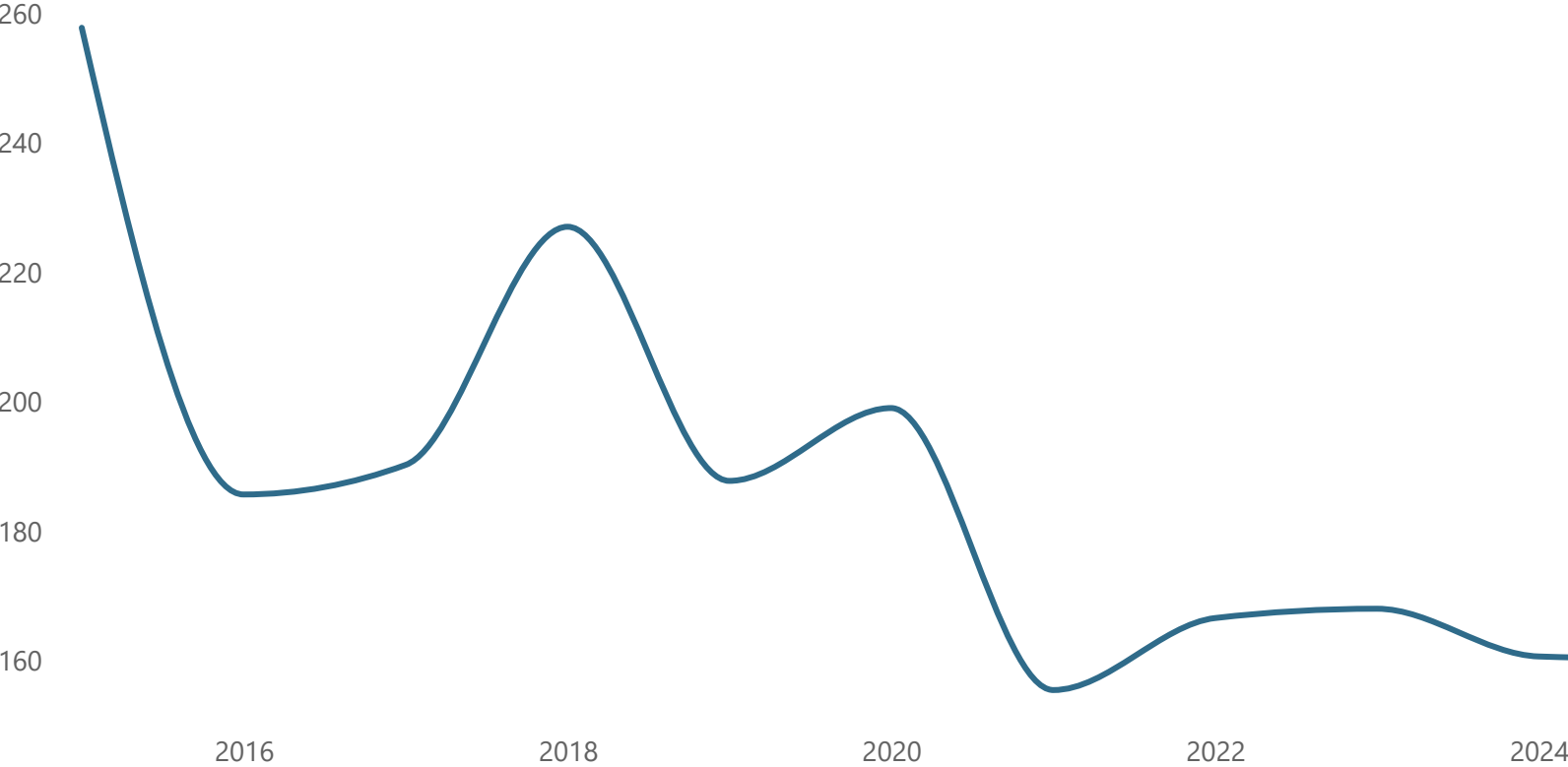
CO₂ Emissions Trend (MtCO₂)



CO₂ Emissions by Energy Source (MtCO₂)



Carbon Intensity Trend (gCO₂/kWh)



Carbon Intensity Change

-37,8 %

CO₂ Emissions Change

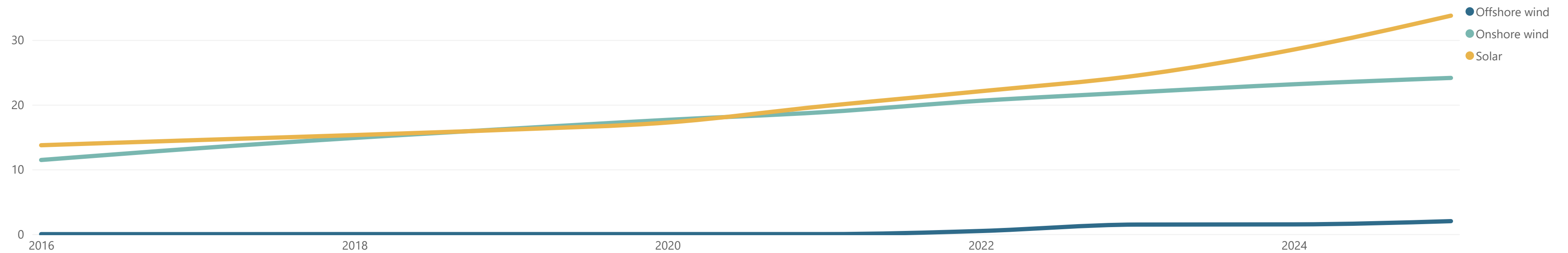
-35,2 %

2015 → 2025

2021

France

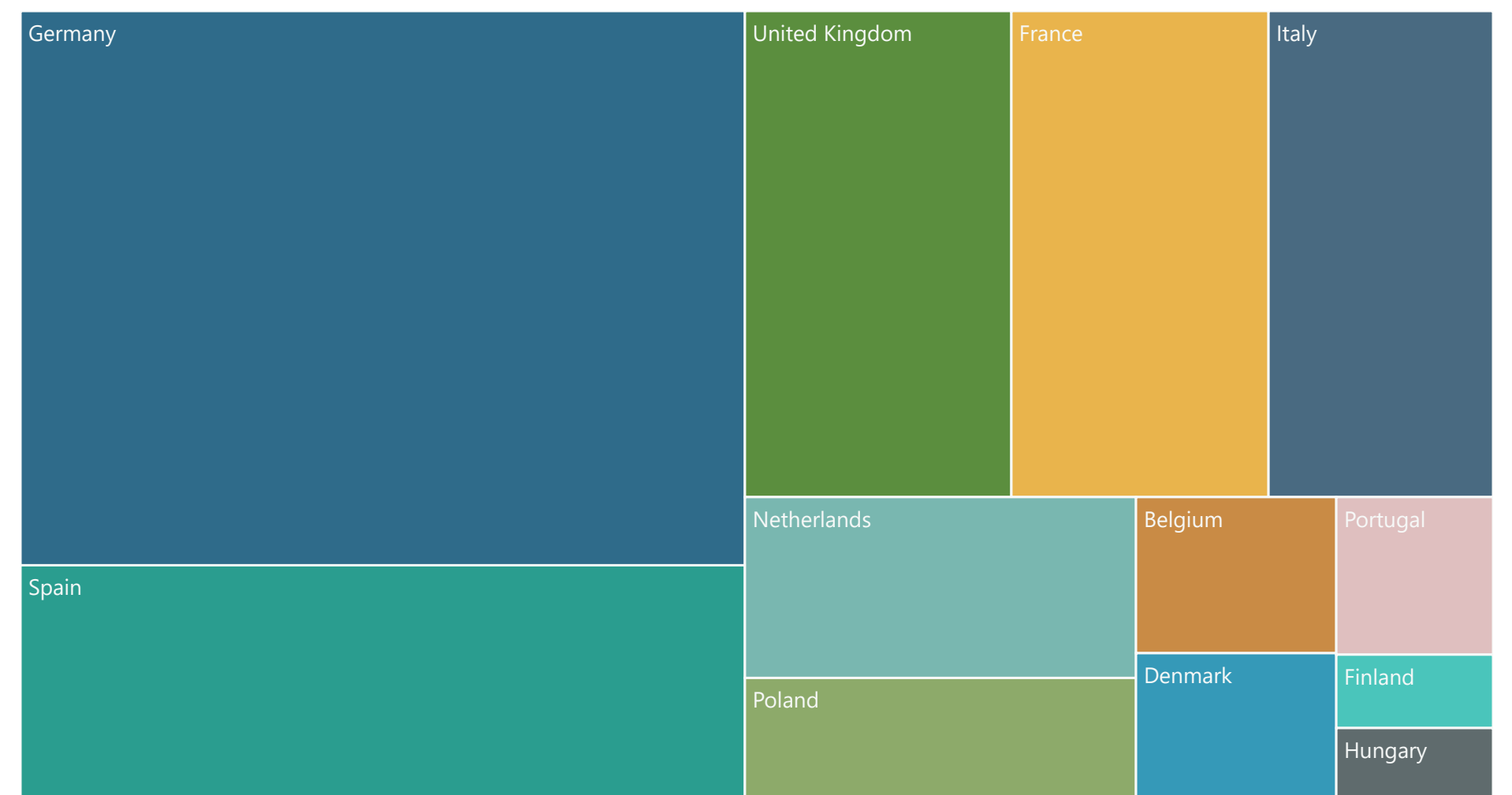
Installed Capacity Trend by Technology (GW)



Installed Capacity by Technology (GW)



Installed Capacity by Country (GW)



Key Insights & Conclusions

01 - Generation & Demand

Electricity generation and demand developed unevenly across the 20 analysed countries between 2017 and 2025. Domestic generation increased in 12 countries and declined in 8, while electricity demand increased in 10 and decreased in 10. Changes in net electricity imports and exports helped bridge the gap between domestic generation and demand, highlighting substantial differences in national electricity-system developments.

02 - Energy Mix & Transition

The electricity generation mix shifted substantially towards renewables between 2017 and 2025. The renewable share increased in all 20 analysed countries, while fossil generation shares generally declined. The scale of the transition varied considerably across countries, with particularly large renewable-share gains in Portugal, the Netherlands, Hungary, the United Kingdom and Germany.

03 - Emissions & Carbon Intensity

Power-sector decarbonisation progressed across the analysed European countries between 2017 and 2025. Carbon intensity declined in all 20 countries, while absolute power-sector CO₂ emissions fell in 19 of 20. The scale of the decline varied considerably, with particularly large emission reductions in Portugal, Finland, Spain and Germany. Switzerland was the only country with higher absolute emissions in 2025, despite a slight decline in carbon intensity.

04 - Installed Capacity

Installed solar and wind capacity expanded substantially between 2016 and 2025 across all nine countries with complete comparable data. Growth was particularly strong in Hungary, the Netherlands and Poland in relative terms, while Germany recorded the largest combined solar and wind capacity in 2025. Solar expansion was the main driver of growth in several countries, although wind capacity also increased substantially in markets such as the Netherlands, France and the United Kingdom.

01 – Generation & Demand: Comparison based on the common complete period 2017–2025. Changes in net imports and exports are considered when interpreting differences between domestic generation and demand.
02 – Energy Mix & Transition: Comparison based on the common complete period 2017–2025. Energy-mix shares exclude net imports and aggregate series to avoid double counting.
03 – Emissions & Carbon Intensity: Comparison based on the common complete period 2017–2025. Total emissions refer to the aggregate “Total generation” emissions series; carbon intensity represents average monthly gCO₂/kWh.
04 – Installed Capacity: Comparison covers 2016–2025 and the nine countries with complete comparable Solar and Wind capacity data. Capacity values represent annual averages of monthly installed capacity.